# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 2-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 383 945 680 437 | 30 560 138 315 | 30 377 509 662 | 181 932 059 | 696 594 | 99.4 | 21.33 | 0.09 | 15.768 | 64.32 | 4.459 | 3.897 | 0.124 | 0.149 |
| parest | 299 200 615 391 | 580 499 665 | 94 157 487 | 462 351 877 | 23 990 301 | 16.2 | 0.18 | 0.31 | 11.513 | 61.90 | 4.712 | 2.954 | 2.226 | 2.168 |
| deepsjeng | 399 987 651 203 | 14 214 822 378 | 3 538 346 146 | 10 408 044 378 | 268 431 854 | 24.9 | 2.99 | 5.20 | 9.802 | 6.58 | 2.937 | 7.867 | 2.873 | 3.816 |
| leela | 436 316 202 851 | 13 338 709 984 | 512 185 059 | 12 317 582 149 | 508 942 776 | 3.8 | 0.43 | 5.65 | 6.412 | 16.65 | 7.022 | 0.201 | 1.846 | 0.874 |
| namd | 218 427 292 248 | 2 350 200 000 | 2 339 066 714 | 8 976 130 | 2 157 156 | 99.5 | 3.22 | 0.01 | 12.792 | 25.42 | 5.330 | 1.389 | 4.247 | 1.170 |
| lbm | 100 160 472 568 | 4 229 647 076 | 4 229 634 900 | 11 332 | 844 | 100.0 | 9.69 | 0.00 | 23.364 | 88.88 | 0.424 | 2.108 | 10.784 | 0.891 |
| nab | 38 232 521 997 | 784 372 830 | 637 890 733 | 128 504 979 | 17 977 118 | 81.3 | 7.61 | 0.67 | 7.239 | 4.20 | 0.995 | 0.807 | 1.159 | 1.446 |
| nab_gcn4dna | 366 315 945 523 | 7 770 025 698 | 5 788 787 408 | 1 746 098 454 | 235 139 836 | 74.5 | 7.30 | 0.95 | 6.931 | 41.67 | 1.143 | 1.378 | 1.260 | 1.841 |
| xz | 45 071 901 345 | 1 313 479 362 | 149 252 420 | 1 077 227 617 | 86 999 325 | 11.4 | 1.36 | 4.78 | 7.110 | 32.18 | 1.406 | 9.928 | 0.562 | 2.003 |
| xz_combined | 103 539 847 907 | 1 438 295 106 | 686 344 414 | 673 142 677 | 78 808 015 | 47.7 | 2.79 | 1.30 | 10.851 | 20.84 | 3.701 | 2.298 | 3.716 | 4.200 |
| imagick | 284 114 941 059 | 294 947 574 | 3 377 502 | 291 114 095 | 455 977 | 1.1 | 0.04 | 0.20 | 5.021 | 94.86 | 34.988 | 0.470 | 0.076 | 0.042 |
| blender | 721 231 649 259 | 11 980 515 650 | 494 531 875 | 10 340 415 008 | 1 145 568 767 | 4.1 | 0.31 | 2.87 | 7.420 | 11.64 | 3.841 | 7.013 | 3.816 | 3.949 |
| povray | 28 621 075 427 | 2 169 255 652 | 27 550 868 | 2 002 183 119 | 139 521 665 | 1.3 | 0.27 | 13.99 | 10.102 | 41.87 | 7.634 | 20.369 | 1.987 | 2.166 |
| omnetpp | 134 781 952 422 | 8 710 005 757 | 5 235 206 | 8 476 934 143 | 227 836 408 | 0.1 | 0.01 | 12.58 | 16.549 | 28.83 | 0.250 | 0.873 | 3.586 | 3.090 |
| xalancbmk | 270 346 763 301 | 25 872 371 471 | 370 722 439 | 24 133 124 322 | 1 368 524 710 | 1.4 | 0.44 | 17.85 | 12.401 | 12.36 | 20.478 | 29.684 | 2.009 | 3.442 |
| x264 | 386 397 511 704 | 16 699 463 061 | 11 425 519 865 | 4 723 224 155 | 550 719 041 | 68.4 | 6.30 | 2.44 | 2.327 | 28.09 | 0.391 | 0.945 | 1.291 | 1.441 |
| gcc_200 | 118 445 469 526 | 9 732 539 869 | 435 131 547 | 9 187 999 599 | 109 408 723 | 4.5 | 1.16 | 15.51 | 9.558 | 36.44 | 2.260 | 5.111 | 0.947 | 1.775 |
| gcc_scilab | 92 316 903 709 | 7 134 405 901 | 365 548 253 | 6 690 997 652 | 77 859 996 | 5.1 | 1.25 | 14.50 | 10.166 | 36.72 | 2.616 | 4.684 | 0.943 | 1.860 |
| gcc | 10 027 112 139 | 797 875 479 | 42 973 169 | 745 789 454 | 9 112 856 | 5.4 | 1.41 | 14.88 | 9.938 | 38.95 | 2.414 | 4.845 | 0.924 | 1.921 |
| mcf | 125 387 999 079 | 1 200 460 284 | 483 048 594 | 714 151 251 | 3 260 439 | 40.2 | 1.61 | 1.14 | 23.825 | 75.08 | 7.037 | 8.489 | 3.741 | 5.922 |
| perlbench_scrabbl | 29 794 284 033 | 2 637 314 691 | 19 766 463 | 2 595 400 347 | 22 147 881 | 0.7 | 0.23 | 17.42 | 7.971 | 11.62 | 3.139 | 11.629 | 1.532 | 1.850 |
| perlbench_suns | 1 489 955 308 | 141 093 681 | 4 683 473 | 133 092 545 | 3 317 663 | 3.3 | 0.98 | 17.87 | 6.547 | 18.32 | 1.109 | 8.999 | 0.638 | 1.128 |
| perlbench_diffmail | 71 841 738 045 | 5 980 785 368 | 357 324 444 | 5 559 068 861 | 64 392 063 | 6.0 | 2.06 | 15.48 | 6.459 | 32.33 | 4.867 | 4.168 | 0.433 | 1.420 |
| perlbench_perfect | 12 407 457 725 | 1 157 150 004 | 57 361 433 | 1 079 863 927 | 19 924 644 | 5.0 | 1.44 | 17.41 | 9.270 | 51.16 | 1.241 | 5.864 | 0.658 | 1.422 |
| perlbench_splitmail | 36 465 452 492 | 2 676 936 460 | 458 223 604 | 2 146 373 965 | 72 338 891 | 17.1 | 5.79 | 11.77 | 5.379 | 17.25 | 1.788 | 2.776 | 1.168 | 0.882 |
| bwaves_1 | 94 269 943 053 | 669 560 992 | 669 492 761 | 55 690 | 12 541 | 100.0 | 2.99 | 0.00 | 42.240 | 27.05 | 3.276 | 0.266 | 4.574 | 5.163 |
| bwaves_2 | 91 219 148 006 | 659 168 858 | 659 100 916 | 55 475 | 12 467 | 100.0 | 3.04 | 0.00 | 42.111 | 27.48 | 3.213 | 0.270 | 4.496 | 5.146 |
| exchange2 | 177 369 845 254 | 5 445 875 346 | 3 838 435 663 | 1 053 305 830 | 554 133 853 | 70.5 | 4.84 | 1.19 | 2.265 | 1.44 | 2.049 | 2.700 | 5.466 | 4.162 |
| fotonik3d | 188 709 212 304 | 1 778 874 318 | 1 763 902 508 | 13 601 309 | 1 370 501 | 99.2 | 3.99 | 0.01 | 17.946 | 38.50 | 5.705 | 0.057 | 0.155 | 3.923 |
| wrf | 519 187 619 962 | 2 589 057 857 | 681 696 831 | 1 750 897 417 | 156 463 609 | 26.3 | 0.92 | 0.67 | 17.589 | 22.33 | 9.173 | 6.100 | 0.861 | 1.685 |
| cam4 | 63 971 148 446 | 1 115 718 661 | 416 171 917 | 673 858 888 | 25 687 856 | 37.3 | 2.49 | 2.11 | 11.199 | 54.29 | 2.588 | 2.625 | 1.073 | 1.861 |
| **Total** | 5 849 595 313 723 | 186 023 567 348 | **70 932 974 274** | 109 315 378 704 | 5 775 214 370 | 38.1 | 3.60 | 3.74 | 0.000 | — | — | — | — | — |

# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, fully associative L1D/L2 (FALRU) · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 383 945 680 437 | 30 560 138 315 | 30 377 509 662 | 181 932 059 | 696 594 | 99.4 | 21.33 | 0.09 | 15.230 | 59.95 | 4.847 | 5.149 | 0.076 | 0.107 |
| parest | 299 200 615 391 | 580 499 665 | 94 157 487 | 462 351 877 | 23 990 301 | 16.2 | 0.18 | 0.31 | 9.868 | 72.07 | 1.057 | 1.252 | 0.316 | 0.589 |
| deepsjeng | 399 987 651 203 | 14 214 822 378 | 3 538 346 146 | 10 408 044 378 | 268 431 854 | 24.9 | 2.99 | 5.20 | 7.787 | 6.05 | 1.446 | 7.512 | 1.630 | 3.365 |
| leela | 436 316 202 851 | 13 338 709 984 | 512 185 059 | 12 317 582 149 | 508 942 776 | 3.8 | 0.43 | 5.65 | 4.486 | 22.42 | 0.198 | 0.199 | 0.834 | 0.260 |
| namd | 218 427 292 248 | 2 350 200 000 | 2 339 066 714 | 8 976 130 | 2 157 156 | 99.5 | 3.22 | 0.01 | 11.492 | 27.80 | 2.422 | 1.263 | 4.202 | 1.058 |
| lbm | 100 160 472 568 | 4 229 647 076 | 4 229 634 900 | 11 332 | 844 | 100.0 | 9.69 | 0.00 | 20.862 | 99.62 | 0.000 | 0.375 | 10.819 | 0.644 |
| nab | 38 232 521 997 | 784 372 830 | 637 890 733 | 128 504 979 | 17 977 118 | 81.3 | 7.61 | 0.67 | 3.868 | 7.20 | 0.089 | 0.172 | 0.449 | 0.973 |
| nab_gcn4dna | 366 315 945 523 | 7 770 025 698 | 5 788 787 408 | 1 746 098 454 | 235 139 836 | 74.5 | 7.30 | 0.95 | 3.911 | 74.36 | 0.062 | 0.214 | 0.746 | 0.948 |
| xz | 45 071 901 345 | 1 313 479 362 | 149 252 420 | 1 077 227 617 | 86 999 325 | 11.4 | 1.36 | 4.78 | 4.769 | 47.29 | 0.029 | 0.695 | 0.082 | 0.030 |
| xz_combined | 103 539 847 907 | 1 438 295 106 | 686 344 414 | 673 142 677 | 78 808 015 | 47.7 | 2.79 | 1.30 | 9.786 | 22.72 | 0.838 | 0.506 | 1.881 | 1.671 |
| imagick | 284 114 941 975 | 294 947 576 | 3 377 502 | 291 114 097 | 455 977 | 1.1 | 0.04 | 0.20 | 5.015 | 94.98 | 34.952 | 0.402 | 0.075 | 0.041 |
| blender | 721 231 649 060 | 11 980 515 650 | 494 531 875 | 10 340 415 008 | 1 145 568 767 | 4.1 | 0.31 | 2.87 | 6.698 | 11.68 | 4.769 | 2.346 | 1.980 | 3.663 |
| povray | 28 621 075 368 | 2 169 255 656 | 27 550 868 | 2 002 183 123 | 139 521 665 | 1.3 | 0.27 | 13.99 | 8.600 | 51.48 | 7.072 | 19.315 | 1.660 | 0.931 |
| omnetpp | 134 781 950 948 | 8 710 005 719 | 5 235 206 | 8 476 934 111 | 227 836 402 | 0.1 | 0.01 | 12.58 | 16.026 | 28.39 | 0.054 | 0.787 | 2.025 | 3.209 |
| xalancbmk | 270 346 763 301 | 25 872 371 471 | 370 722 439 | 24 133 124 322 | 1 368 524 710 | 1.4 | 0.44 | 17.85 | 11.445 | 6.67 | 6.735 | 32.171 | 1.595 | 3.835 |
| x264 | 386 397 511 704 | 16 699 463 061 | 11 425 519 865 | 4 723 224 155 | 550 719 041 | 68.4 | 6.30 | 2.44 | 1.802 | 33.73 | 0.149 | 0.826 | 0.803 | 1.025 |
| gcc_200 | 118 445 469 546 | 9 732 539 869 | 435 131 547 | 9 187 999 599 | 109 408 723 | 4.5 | 1.16 | 15.51 | 7.531 | 43.03 | 0.808 | 4.066 | 0.443 | 1.197 |
| gcc_scilab | 92 316 903 709 | 7 134 405 901 | 365 548 253 | 6 690 997 652 | 77 859 996 | 5.1 | 1.25 | 14.50 | 8.134 | 42.90 | 1.011 | 3.830 | 0.454 | 1.275 |
| gcc | 10 027 112 139 | 797 875 479 | 42 973 169 | 745 789 454 | 9 112 856 | 5.4 | 1.41 | 14.88 | 7.800 | 46.27 | 0.736 | 4.106 | 0.405 | 1.274 |
| mcf | 125 387 999 079 | 1 200 460 284 | 483 048 594 | 714 151 251 | 3 260 439 | 40.2 | 1.61 | 1.14 | 22.957 | 77.43 | 3.669 | 7.069 | 2.618 | 5.400 |
| perlbench_scrabbl | 29 794 284 033 | 2 637 314 691 | 19 766 463 | 2 595 400 347 | 22 147 881 | 0.7 | 0.23 | 17.42 | 4.936 | 6.46 | 5.076 | 5.738 | 0.209 | 0.636 |
| perlbench_suns | 1 489 955 308 | 141 093 681 | 4 683 473 | 133 092 545 | 3 317 663 | 3.3 | 0.98 | 17.87 | 4.620 | 20.06 | 0.127 | 3.311 | 0.052 | 0.568 |
| perlbench_diffmail | 71 841 738 045 | 5 980 785 368 | 357 324 444 | 5 559 068 861 | 64 392 063 | 6.0 | 2.06 | 15.48 | 3.030 | 53.90 | 0.146 | 2.236 | 0.119 | 0.402 |
| perlbench_perfect | 12 407 457 725 | 1 157 150 004 | 57 361 433 | 1 079 863 927 | 19 924 644 | 5.0 | 1.44 | 17.41 | 7.516 | 56.72 | 0.454 | 5.096 | 0.173 | 0.751 |
| perlbench_splitmail | 36 465 452 492 | 2 676 936 460 | 458 223 604 | 2 146 373 965 | 72 338 891 | 17.1 | 5.79 | 11.77 | 3.491 | 20.52 | 0.049 | 1.869 | 0.121 | 0.719 |
| bwaves_1 | 94 269 943 053 | 669 560 992 | 669 492 761 | 55 690 | 12 541 | 100.0 | 2.99 | 0.00 | 7.286 | 97.31 | 0.738 | 0.000 | 3.886 | 5.022 |
| bwaves_2 | 91 219 148 006 | 659 168 858 | 659 100 916 | 55 475 | 12 467 | 100.0 | 3.04 | 0.00 | 7.286 | 97.23 | 0.740 | 0.000 | 3.807 | 5.006 |
| exchange2 | 177 369 845 254 | 5 445 875 346 | 3 838 435 663 | 1 053 305 830 | 554 133 853 | 70.5 | 4.84 | 1.19 | 1.317 | 2.51 | 1.285 | 1.879 | 2.323 | 1.802 |
| fotonik3d | 188 709 212 304 | 1 778 874 318 | 1 763 902 508 | 13 601 309 | 1 370 501 | 99.2 | 3.99 | 0.01 | 8.391 | 82.20 | 1.988 | 0.006 | 0.091 | 3.864 |
| wrf | 519 187 576 810 | 2 589 056 629 | 681 696 831 | 1 750 896 547 | 156 463 251 | 26.3 | 0.92 | 0.67 | 5.784 | 61.40 | 5.291 | 4.181 | 0.452 | 1.323 |
| cam4 | 63 971 148 409 | 1 115 718 655 | 416 171 917 | 673 858 883 | 25 687 855 | 37.3 | 2.49 | 2.11 | 9.806 | 60.40 | 1.479 | 1.854 | 0.595 | 1.405 |
| **Total** | 5 849 595 269 738 | 186 023 566 082 | **70 932 974 274** | 109 315 377 803 | 5 775 214 005 | 38.1 | 3.60 | 3.74 | 0.000 | — | — | — | — | — |

# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, direct-mapped L1D/L2 · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 383 945 680 437 | 30 560 138 315 | 30 377 509 662 | 181 932 059 | 696 594 | 99.4 | 21.33 | 0.09 | 17.365 | 48.81 | 6.420 | 5.496 | 1.484 | 0.271 |
| parest | 299 200 615 391 | 580 499 665 | 94 157 487 | 462 351 877 | 23 990 301 | 16.2 | 0.18 | 0.31 | 15.291 | 50.52 | 13.828 | 5.544 | 4.604 | 5.835 |
| deepsjeng | 399 987 651 203 | 14 214 822 378 | 3 538 346 146 | 10 408 044 378 | 268 431 854 | 24.9 | 2.99 | 5.20 | 13.572 | 20.58 | 6.641 | 9.693 | 3.991 | 4.802 |
| leela | 436 316 202 851 | 13 338 709 984 | 512 185 059 | 12 317 582 149 | 508 942 776 | 3.8 | 0.43 | 5.65 | 7.771 | 15.93 | 12.088 | 0.225 | 3.153 | 1.900 |
| namd | 218 427 292 248 | 2 350 200 000 | 2 339 066 714 | 8 976 130 | 2 157 156 | 99.5 | 3.22 | 0.01 | 15.525 | 37.27 | 11.949 | 2.502 | 4.394 | 2.397 |
| lbm | 100 160 472 568 | 4 229 647 076 | 4 229 634 900 | 11 332 | 844 | 100.0 | 9.69 | 0.00 | 26.827 | 74.99 | 1.629 | 5.336 | 10.678 | 1.209 |
| nab | 38 232 521 997 | 784 372 830 | 637 890 733 | 128 504 979 | 17 977 118 | 81.3 | 7.61 | 0.67 | 11.611 | 16.55 | 4.174 | 3.096 | 3.381 | 3.202 |
| nab_gcn4dna | 366 315 945 523 | 7 770 025 698 | 5 788 787 408 | 1 746 098 454 | 235 139 836 | 74.5 | 7.30 | 0.95 | 11.338 | 31.28 | 4.327 | 3.986 | 3.149 | 3.343 |
| xz | 45 071 901 345 | 1 313 479 362 | 149 252 420 | 1 077 227 617 | 86 999 325 | 11.4 | 1.36 | 4.78 | 8.854 | 32.06 | 3.589 | 9.751 | 1.675 | 2.574 |
| xz_combined | 103 539 847 907 | 1 438 295 106 | 686 344 414 | 673 142 677 | 78 808 015 | 47.7 | 2.79 | 1.30 | 14.676 | 25.02 | 7.094 | 5.232 | 5.413 | 6.019 |
| imagick | 284 114 941 975 | 294 947 576 | 3 377 502 | 291 114 097 | 455 977 | 1.1 | 0.04 | 0.20 | 5.624 | 86.53 | 28.947 | 2.476 | 0.090 | 0.304 |
| blender | 721 231 649 060 | 11 980 515 650 | 494 531 875 | 10 340 415 008 | 1 145 568 767 | 4.1 | 0.31 | 2.87 | 9.957 | 27.63 | 6.675 | 7.114 | 5.582 | 6.198 |
| povray | 28 621 640 749 | 2 169 279 970 | 27 550 868 | 2 002 206 555 | 139 522 547 | 1.3 | 0.27 | 13.99 | 14.296 | 36.10 | 12.024 | 21.547 | 4.564 | 4.442 |
| omnetpp | 134 781 950 948 | 8 710 005 719 | 5 235 206 | 8 476 934 111 | 227 836 402 | 0.1 | 0.01 | 12.58 | 18.539 | 41.67 | 0.723 | 10.689 | 4.354 | 4.020 |
| xalancbmk | 270 346 763 301 | 25 872 371 471 | 370 722 439 | 24 133 124 322 | 1 368 524 710 | 1.4 | 0.44 | 17.85 | 14.578 | 26.71 | 30.232 | 25.465 | 3.104 | 4.949 |
| x264 | 386 397 511 704 | 16 699 463 061 | 11 425 519 865 | 4 723 224 155 | 550 719 041 | 68.4 | 6.30 | 2.44 | 4.045 | 26.31 | 0.928 | 1.200 | 1.720 | 2.173 |
| gcc_200 | 118 445 469 526 | 9 732 539 869 | 435 131 547 | 9 187 999 599 | 109 408 723 | 4.5 | 1.16 | 15.51 | 12.949 | 37.34 | 7.301 | 8.798 | 2.693 | 3.571 |
| gcc_scilab | 92 316 903 709 | 7 134 405 901 | 365 548 253 | 6 690 997 652 | 77 859 996 | 5.1 | 1.25 | 14.50 | 13.633 | 38.00 | 7.843 | 8.196 | 2.770 | 3.710 |
| gcc | 10 027 112 139 | 797 875 479 | 42 973 169 | 745 789 454 | 9 112 856 | 5.4 | 1.41 | 14.88 | 13.466 | 39.11 | 6.700 | 8.236 | 2.821 | 3.814 |
| mcf | 125 387 999 079 | 1 200 460 284 | 483 048 594 | 714 151 251 | 3 260 439 | 40.2 | 1.61 | 1.14 | 26.678 | 71.66 | 10.486 | 10.536 | 5.620 | 6.519 |
| perlbench_scrabbl | 29 794 284 033 | 2 637 314 691 | 19 766 463 | 2 595 400 347 | 22 147 881 | 0.7 | 0.23 | 17.42 | 12.387 | 16.92 | 5.421 | 15.503 | 7.057 | 4.858 |
| perlbench_suns | 1 489 955 308 | 141 093 681 | 4 683 473 | 133 092 545 | 3 317 663 | 3.3 | 0.98 | 17.87 | 9.862 | 25.17 | 2.953 | 10.611 | 3.530 | 2.782 |
| perlbench_diffmail | 71 841 738 045 | 5 980 785 368 | 357 324 444 | 5 559 068 861 | 64 392 063 | 6.0 | 2.06 | 15.48 | 10.331 | 26.66 | 2.391 | 10.536 | 3.199 | 3.979 |
| perlbench_perfect | 12 407 457 725 | 1 157 150 004 | 57 361 433 | 1 079 863 927 | 19 924 644 | 5.0 | 1.44 | 17.41 | 11.060 | 49.94 | 1.540 | 7.692 | 1.660 | 2.386 |
| perlbench_splitmail | 36 465 452 492 | 2 676 936 460 | 458 223 604 | 2 146 373 965 | 72 338 891 | 17.1 | 5.79 | 11.77 | 8.886 | 21.52 | 3.081 | 2.969 | 3.127 | 2.272 |
| bwaves_1 | 94 269 943 053 | 669 560 992 | 669 492 761 | 55 690 | 12 541 | 100.0 | 2.99 | 0.00 | 43.947 | 26.39 | 6.178 | 1.303 | 4.893 | 5.680 |
| bwaves_2 | 91 219 148 006 | 659 168 858 | 659 100 916 | 55 475 | 12 467 | 100.0 | 3.04 | 0.00 | 43.905 | 26.65 | 6.082 | 1.324 | 4.827 | 5.666 |
| exchange2 | 177 369 845 254 | 5 445 875 346 | 3 838 435 663 | 1 053 305 830 | 554 133 853 | 70.5 | 4.84 | 1.19 | 3.655 | 1.48 | 2.783 | 2.443 | 2.474 | 3.955 |
| fotonik3d | 188 709 212 304 | 1 778 874 318 | 1 763 902 508 | 13 601 309 | 1 370 501 | 99.2 | 3.99 | 0.01 | 22.468 | 36.82 | 9.882 | 0.425 | 0.760 | 3.970 |
| wrf | 519 187 794 216 | 2 589 068 129 | 681 696 831 | 1 750 905 769 | 156 465 529 | 26.3 | 0.92 | 0.67 | 25.818 | 26.35 | 13.717 | 8.463 | 5.685 | 2.616 |
| cam4 | 63 971 148 372 | 1 115 718 649 | 416 171 917 | 673 858 878 | 25 687 854 | 37.3 | 2.49 | 2.11 | 14.713 | 46.92 | 5.035 | 3.661 | 2.137 | 2.781 |
| **Total** | 5 849 596 052 468 | 186 023 601 890 | **70 932 974 274** | 109 315 410 452 | 5 775 217 164 | 38.1 | 3.60 | 3.74 | 0.000 | — | — | — | — | — |

# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 4-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 383 945 680 437 | 30 560 138 315 | 30 377 509 662 | 181 932 059 | 696 594 | 99.4 | 21.33 | 0.09 | 15.301 | 65.68 | 4.854 | 4.273 | 0.082 | 0.112 |
| parest | 299 200 615 391 | 580 499 665 | 94 157 487 | 462 351 877 | 23 990 301 | 16.2 | 0.18 | 0.31 | 10.271 | 69.50 | 2.031 | 1.458 | 0.507 | 1.069 |
| deepsjeng | 399 987 651 203 | 14 214 822 378 | 3 538 346 146 | 10 408 044 378 | 268 431 854 | 24.9 | 2.99 | 5.20 | 8.712 | 7.25 | 2.022 | 7.465 | 2.162 | 3.395 |
| leela | 436 316 202 851 | 13 338 709 984 | 512 185 059 | 12 317 582 149 | 508 942 776 | 3.8 | 0.43 | 5.65 | 5.259 | 20.07 | 1.739 | 0.199 | 1.591 | 0.409 |
| namd | 218 427 292 248 | 2 350 200 000 | 2 339 066 714 | 8 976 130 | 2 157 156 | 99.5 | 3.22 | 0.01 | 11.587 | 28.04 | 2.638 | 1.229 | 4.226 | 1.077 |
| lbm | 100 160 472 568 | 4 229 647 076 | 4 229 634 900 | 11 332 | 844 | 100.0 | 9.69 | 0.00 | 21.300 | 97.61 | 0.000 | 0.504 | 10.784 | 0.679 |
| nab | 38 232 521 997 | 784 372 830 | 637 890 733 | 128 504 979 | 17 977 118 | 81.3 | 7.61 | 0.67 | 5.477 | 5.51 | 0.418 | 0.274 | 0.502 | 0.931 |
| nab_gcn4dna | 366 315 945 523 | 7 770 025 698 | 5 788 787 408 | 1 746 098 454 | 235 139 836 | 74.5 | 7.30 | 0.95 | 4.915 | 58.63 | 0.236 | 0.346 | 0.775 | 1.183 |
| xz | 45 071 901 345 | 1 313 479 362 | 149 252 420 | 1 077 227 617 | 86 999 325 | 11.4 | 1.36 | 4.78 | 5.998 | 38.14 | 0.870 | 2.994 | 0.156 | 0.476 |
| xz_combined | 103 539 847 907 | 1 438 295 106 | 686 344 414 | 673 142 677 | 78 808 015 | 47.7 | 2.79 | 1.30 | 9.823 | 23.02 | 1.748 | 0.813 | 2.573 | 2.475 |
| imagick | 284 114 941 975 | 294 947 576 | 3 377 502 | 291 114 097 | 455 977 | 1.1 | 0.04 | 0.20 | 5.014 | 94.98 | 34.952 | 0.476 | 0.075 | 0.041 |
| blender | 721 231 649 259 | 11 980 515 650 | 494 531 875 | 10 340 415 008 | 1 145 568 767 | 4.1 | 0.31 | 2.87 | 6.708 | 12.80 | 5.329 | 4.780 | 2.492 | 3.670 |
| povray | 28 621 035 222 | 2 169 253 915 | 27 550 868 | 2 002 181 445 | 139 521 602 | 1.3 | 0.27 | 13.99 | 9.021 | 46.64 | 6.986 | 19.463 | 1.764 | 1.141 |
| omnetpp | 134 781 950 948 | 8 710 005 719 | 5 235 206 | 8 476 934 111 | 227 836 402 | 0.1 | 0.01 | 12.58 | 15.943 | 29.81 | 0.054 | 0.832 | 2.484 | 2.696 |
| xalancbmk | 270 346 763 301 | 25 872 371 471 | 370 722 439 | 24 133 124 322 | 1 368 524 710 | 1.4 | 0.44 | 17.85 | 11.879 | 12.81 | 15.327 | 26.563 | 1.679 | 3.436 |
| x264 | 386 397 511 704 | 16 699 463 061 | 11 425 519 865 | 4 723 224 155 | 550 719 041 | 68.4 | 6.30 | 2.44 | 1.969 | 32.24 | 0.220 | 0.877 | 0.856 | 1.155 |
| gcc_200 | 118 445 469 526 | 9 732 539 869 | 435 131 547 | 9 187 999 599 | 109 408 723 | 4.5 | 1.16 | 15.51 | 8.321 | 41.08 | 1.205 | 4.233 | 0.560 | 1.336 |
| gcc_scilab | 92 316 903 709 | 7 134 405 901 | 365 548 253 | 6 690 997 652 | 77 859 996 | 5.1 | 1.25 | 14.50 | 8.897 | 41.20 | 1.479 | 3.912 | 0.554 | 1.412 |
| gcc | 10 027 112 139 | 797 875 479 | 42 973 169 | 745 789 454 | 9 112 856 | 5.4 | 1.41 | 14.88 | 8.690 | 43.73 | 1.461 | 4.121 | 0.519 | 1.449 |
| mcf | 125 387 999 079 | 1 200 460 284 | 483 048 594 | 714 151 251 | 3 260 439 | 40.2 | 1.61 | 1.14 | 23.171 | 77.15 | 5.261 | 7.394 | 2.821 | 5.569 |
| perlbench_scrabbl | 29 794 284 033 | 2 637 314 691 | 19 766 463 | 2 595 400 347 | 22 147 881 | 0.7 | 0.23 | 17.42 | 6.200 | 14.10 | 3.288 | 8.908 | 0.265 | 0.682 |
| perlbench_suns | 1 489 955 308 | 141 093 681 | 4 683 473 | 133 092 545 | 3 317 663 | 3.3 | 0.98 | 17.87 | 5.444 | 20.70 | 0.394 | 8.944 | 0.204 | 0.625 |
| perlbench_diffmail | 71 841 738 045 | 5 980 785 368 | 357 324 444 | 5 559 068 861 | 64 392 063 | 6.0 | 2.06 | 15.48 | 4.666 | 41.21 | 0.160 | 2.757 | 0.172 | 0.737 |
| perlbench_perfect | 12 407 457 725 | 1 157 150 004 | 57 361 433 | 1 079 863 927 | 19 924 644 | 5.0 | 1.44 | 17.41 | 8.248 | 55.17 | 0.675 | 5.518 | 0.222 | 1.087 |
| perlbench_splitmail | 36 465 452 492 | 2 676 936 460 | 458 223 604 | 2 146 373 965 | 72 338 891 | 17.1 | 5.79 | 11.77 | 4.500 | 19.98 | 0.167 | 2.964 | 0.234 | 0.575 |
| bwaves_1 | 94 269 943 053 | 669 560 992 | 669 492 761 | 55 690 | 12 541 | 100.0 | 2.99 | 0.00 | 43.002 | 26.58 | 4.978 | 0.139 | 3.846 | 4.943 |
| bwaves_2 | 91 219 148 006 | 659 168 858 | 659 100 916 | 55 475 | 12 467 | 100.0 | 3.04 | 0.00 | 42.856 | 27.02 | 4.852 | 0.141 | 3.762 | 4.927 |
| exchange2 | 177 369 845 254 | 5 445 875 346 | 3 838 435 663 | 1 053 305 830 | 554 133 853 | 70.5 | 4.84 | 1.19 | 1.849 | 1.76 | 1.404 | 2.347 | 4.088 | 1.542 |
| fotonik3d | 188 709 212 304 | 1 778 874 318 | 1 763 902 508 | 13 601 309 | 1 370 501 | 99.2 | 3.99 | 0.01 | 11.668 | 59.27 | 3.812 | 0.008 | 0.090 | 3.852 |
| wrf | 519 187 585 747 | 2 589 056 886 | 681 696 831 | 1 750 896 722 | 156 463 333 | 26.3 | 0.92 | 0.67 | 11.054 | 34.94 | 6.761 | 4.696 | 0.533 | 1.467 |
| cam4 | 63 971 148 409 | 1 115 718 655 | 416 171 917 | 673 858 883 | 25 687 855 | 37.3 | 2.49 | 2.11 | 10.122 | 59.86 | 1.765 | 2.127 | 0.641 | 1.451 |
| **Total** | 5 849 595 238 708 | 186 023 564 598 | **70 932 974 274** | 109 315 376 300 | 5 775 214 024 | 38.1 | 3.60 | 3.74 | 0.000 | — | — | — | — | — |

# Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 8-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

**Definitions (metric schema 9).** Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

| Benchmark | ROI_insts | stack_event | **spill_event** | csave_event | other_event | spill/stack % | Spill % | Csave % | L1D m % | L2 m % | sp_load m % | sp_store m % | cs_load m % | cs_store m % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cactuBSSN | 383 945 680 437 | 30 560 138 315 | 30 377 509 662 | 181 932 059 | 696 594 | 99.4 | 21.33 | 0.09 | 15.629 | 64.55 | 5.021 | 4.672 | 0.081 | 0.113 |
| parest | 299 200 615 391 | 580 499 665 | 94 157 487 | 462 351 877 | 23 990 301 | 16.2 | 0.18 | 0.31 | 10.031 | 71.14 | 1.200 | 1.243 | 0.363 | 0.806 |
| deepsjeng | 399 987 651 203 | 14 214 822 378 | 3 538 346 146 | 10 408 044 378 | 268 431 854 | 24.9 | 2.99 | 5.20 | 8.409 | 7.44 | 1.798 | 7.315 | 1.817 | 3.419 |
| leela | 436 316 202 851 | 13 338 709 984 | 512 185 059 | 12 317 582 149 | 508 942 776 | 3.8 | 0.43 | 5.65 | 4.786 | 21.89 | 0.249 | 0.199 | 1.173 | 0.274 |
| namd | 218 427 292 248 | 2 350 200 000 | 2 339 066 714 | 8 976 130 | 2 157 156 | 99.5 | 3.22 | 0.01 | 11.484 | 28.27 | 2.410 | 1.254 | 4.207 | 1.060 |
| lbm | 100 160 472 568 | 4 229 647 076 | 4 229 634 900 | 11 332 | 844 | 100.0 | 9.69 | 0.00 | 21.366 | 97.30 | 0.000 | 0.460 | 10.748 | 0.715 |
| nab | 38 232 521 997 | 784 372 830 | 637 890 733 | 128 504 979 | 17 977 118 | 81.3 | 7.61 | 0.67 | 4.043 | 7.34 | 0.113 | 0.177 | 0.498 | 0.902 |
| nab_gcn4dna | 366 315 945 523 | 7 770 025 698 | 5 788 787 408 | 1 746 098 454 | 235 139 836 | 74.5 | 7.30 | 0.95 | 4.037 | 71.26 | 0.078 | 0.267 | 0.723 | 1.133 |
| xz | 45 071 901 345 | 1 313 479 362 | 149 252 420 | 1 077 227 617 | 86 999 325 | 11.4 | 1.36 | 4.78 | 5.008 | 45.61 | 0.084 | 1.391 | 0.087 | 0.093 |
| xz_combined | 103 539 847 907 | 1 438 295 106 | 686 344 414 | 673 142 677 | 78 808 015 | 47.7 | 2.79 | 1.30 | 9.704 | 23.25 | 1.016 | 0.741 | 2.038 | 2.103 |
| imagick | 284 114 941 059 | 294 947 574 | 3 377 502 | 291 114 095 | 455 977 | 1.1 | 0.04 | 0.20 | 5.014 | 94.98 | 34.952 | 0.401 | 0.075 | 0.041 |
| blender | 721 231 649 259 | 11 980 515 650 | 494 531 875 | 10 340 415 008 | 1 145 568 767 | 4.1 | 0.31 | 2.87 | 6.674 | 12.85 | 4.894 | 4.398 | 2.202 | 3.547 |
| povray | 28 620 994 980 | 2 169 252 174 | 27 550 868 | 2 002 179 767 | 139 521 539 | 1.3 | 0.27 | 13.99 | 8.577 | 48.92 | 6.967 | 19.272 | 1.517 | 0.966 |
| omnetpp | 134 781 950 948 | 8 710 005 719 | 5 235 206 | 8 476 934 111 | 227 836 402 | 0.1 | 0.01 | 12.58 | 15.902 | 29.84 | 0.053 | 0.732 | 2.206 | 2.863 |
| xalancbmk | 270 346 763 301 | 25 872 371 471 | 370 722 439 | 24 133 124 322 | 1 368 524 710 | 1.4 | 0.44 | 17.85 | 11.396 | 13.19 | 12.128 | 23.991 | 1.593 | 3.761 |
| x264 | 386 397 511 704 | 16 699 463 061 | 11 425 519 865 | 4 723 224 155 | 550 719 041 | 68.4 | 6.30 | 2.44 | 1.849 | 33.89 | 0.151 | 0.850 | 0.808 | 1.078 |
| gcc_200 | 118 445 469 526 | 9 732 539 869 | 435 131 547 | 9 187 999 599 | 109 408 723 | 4.5 | 1.16 | 15.51 | 7.826 | 43.28 | 0.898 | 4.055 | 0.486 | 1.233 |
| gcc_scilab | 92 316 903 709 | 7 134 405 901 | 365 548 253 | 6 690 997 652 | 77 859 996 | 5.1 | 1.25 | 14.50 | 8.426 | 43.14 | 1.148 | 3.779 | 0.490 | 1.299 |
| gcc | 10 027 112 139 | 797 875 479 | 42 973 169 | 745 789 454 | 9 112 856 | 5.4 | 1.41 | 14.88 | 8.166 | 46.10 | 0.977 | 4.007 | 0.442 | 1.310 |
| mcf | 125 387 999 079 | 1 200 460 284 | 483 048 594 | 714 151 251 | 3 260 439 | 40.2 | 1.61 | 1.14 | 22.970 | 77.82 | 4.495 | 7.115 | 2.714 | 5.370 |
| perlbench_scrabbl | 29 794 284 033 | 2 637 314 691 | 19 766 463 | 2 595 400 347 | 22 147 881 | 0.7 | 0.23 | 17.42 | 5.680 | 14.81 | 4.634 | 9.307 | 0.224 | 0.805 |
| perlbench_suns | 1 489 955 308 | 141 093 681 | 4 683 473 | 133 092 545 | 3 317 663 | 3.3 | 0.98 | 17.87 | 4.981 | 22.02 | 0.171 | 5.353 | 0.133 | 0.605 |
| perlbench_diffmail | 71 841 738 045 | 5 980 785 368 | 357 324 444 | 5 559 068 861 | 64 392 063 | 6.0 | 2.06 | 15.48 | 4.489 | 42.24 | 0.128 | 5.789 | 0.110 | 0.755 |
| perlbench_perfect | 12 407 457 725 | 1 157 150 004 | 57 361 433 | 1 079 863 927 | 19 924 644 | 5.0 | 1.44 | 17.41 | 7.714 | 57.56 | 0.632 | 5.224 | 0.179 | 0.899 |
| perlbench_splitmail | 36 465 452 492 | 2 676 936 460 | 458 223 604 | 2 146 373 965 | 72 338 891 | 17.1 | 5.79 | 11.77 | 4.022 | 21.83 | 0.037 | 2.072 | 0.133 | 0.627 |
| bwaves_1 | 94 269 943 053 | 669 560 992 | 669 492 761 | 55 690 | 12 541 | 100.0 | 2.99 | 0.00 | 24.890 | 36.09 | 4.878 | 0.090 | 3.855 | 4.985 |
| bwaves_2 | 91 219 148 006 | 659 168 858 | 659 100 916 | 55 475 | 12 467 | 100.0 | 3.04 | 0.00 | 24.949 | 36.24 | 4.753 | 0.092 | 3.775 | 4.968 |
| exchange2 | 177 369 845 254 | 5 445 875 346 | 3 838 435 663 | 1 053 305 830 | 554 133 853 | 70.5 | 4.84 | 1.19 | 1.082 | 2.98 | 1.018 | 2.036 | 2.370 | 1.737 |
| fotonik3d | 188 709 212 304 | 1 778 874 318 | 1 763 902 508 | 13 601 309 | 1 370 501 | 99.2 | 3.99 | 0.01 | 8.733 | 79.16 | 2.492 | 0.006 | 0.086 | 3.857 |
| wrf | 519 187 577 500 | 2 589 056 652 | 681 696 831 | 1 750 896 547 | 156 463 274 | 26.3 | 0.92 | 0.67 | 6.623 | 56.91 | 6.009 | 4.446 | 0.441 | 1.333 |
| cam4 | 63 971 148 446 | 1 115 718 661 | 416 171 917 | 673 858 888 | 25 687 856 | 37.3 | 2.49 | 2.11 | 9.862 | 61.38 | 1.516 | 1.902 | 0.580 | 1.453 |
| **Total** | 5 849 595 189 340 | 186 023 562 627 | **70 932 974 274** | 109 315 374 450 | 5 775 213 903 | 38.1 | 3.60 | 3.74 | 0.000 | — | — | — | — | — |